

JOSÉ DANIEL GARCÍA CASTRO

// C# / .NET DEVELOPER · MECHANICAL ENGINEER · ENERGY TRANSITION

CONTACT

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Guadalajara, Jalisco, México.
(Open to relocation)

SKILLS

Languages: C#, .NET 8, Python, SQL, G-code
Tools: Git, AutoCAD .NET API, Linux Arch
CAD/CAM: SolidWorks, SW CAM, AutoCAD
Manufacturing: subtractive, additive, laser cutting, molding, thermoforming
Quality: ISO 9001, GD&T, metrology
CNC: Haas, Fanuc, GRBL, Mastercam
IT: servers, networks, ERP, IT support
Process: SDLC end-to-end, Agile, PMBOK
Automotive mechanics: engine, transmission, brakes, steering, diesel and petrol

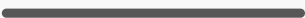
EDUCATION

B.Eng. Mechanical Engineering
ITESO Universidad Jesuita de
Guadalajara
2020 - 2024

Secondary Education (ESO)
Colegio Claret, Madrid
2015 - 2018

High School (Bachillerato)
Colegio Guadalajara
2018 - 2020

LANGUAGES

Spanish (C2) 
English (C1) 
German (A1) 

PROFILE

Mechanical Engineer (graduated Dec. 2024) with a strong background in thermodynamics, energy systems, and heat transfer. Complemented by hands-on experience in industrial software development, precision CNC manufacturing, and metrology. Committed to the energy transition from a systems perspective. Capable of bridging mechanical design, automation, and data analysis to support efficiency and sustainable infrastructure projects. International background with nearly four years in Spain, and with experience collaborating with community centres and NGOs. Brings a track record of operating effectively across disciplines, from writing production software to running 5-axis CNC to managing plant IT infrastructure. Engineers who can only do one thing are common. Engineers who connect all of them are not.

WORK EXPERIENCE

C# Developer — Industrial Software May 2025 – Mar. 2026
Performance Designs / American Industries

- Designed and built an AutoCAD plugin in C# / .NET 8 with a polygon nesting engine for fabric cutting optimization — sole end-to-end developer.
- Owned the full software lifecycle autonomously (requirements, development, testing and maintenance) in a cross-border MX/USA environment with technical and non-technical stakeholders.
- Designed network continuity system (Starlink + UPS failover) and managed plant IT infrastructure. Owned tools roadmap with no senior supervision on-site.

CNC Programmer — Precision Design & Manufacturing
Encore Tools S.A. de C.V. Feb. 2025 – May 2025

- Executed the full CNC manufacturing cycle: read and interpreted GD&T drawings, programmed CAM toolpaths in SolidWorks, generated G-code for the Haas TM2P, and delivered finished aluminum and steel components.
- Validated each part dimensionally using precision metrology instruments (calipers, micrometers), ensuring conformance with design tolerances before release.
- Operated industrial laser cutters as part of the production workflow; managed cutting parameters and surface quality control for multi-gauge steel.

Mechanical Technician / CNC Operator
Moldtech S. de R.L. May 2022 – Aug. 2022 / Aug. 2024 – Dec. 2024

- Gained hands-on experience in precision manufacturing: learned to operate CNC machinery, interpret technical drawings, and apply metrology (calipers, CNC fixtures) in a continuous production environment.
- Developed dimensional quality judgment through geometric tolerance application in production runs, with progressive improvements in changeover times and repeatability.
- Participated in internal engineering projects from design through assembly and machine operation.

CNC Operator
TDI Tecnología & Diseño Industrial May 2023 – Aug. 2023

- Operated 5-axis CNC machinery for medical device manufacturing (intramedullary nails and osteosynthesis plates) under strict quality protocols.
- Applied precision metrology instruments (optical comparators, Vernier calipers, micrometers) for dimensional verification of regulated medical components.
- Managed quality control documentation in standardized formats aligned with ISO and NOM regulations.

Other roles:

- Automotive mechanic: brakes, steering, transmission, drivetrain, suspension.
- Freelance: repair, assembly, and programming of a Satycsa 6174 CNC Router for EntreAmigos Community Center.
- Teleperformance: customer service (100% English) on AT&T campaign.